Python Lists - Short Notes

Software Development I (4COSC006C) · typed up from handwritten notes (March 2023)

What are lists?

- A list is a data type in Python: an ordered sequence of elements, where the elements are **mutable**.
- **Mutable:** the items inside can be changed while the ordered sequence stays, with each item referenced by its index location.
- The first element of a list is always at index 0.
- Each element inside a list is called an **item**. Each item is assigned an index (its location in memory).
- Creating an empty list variable: `mylist = []`
- Lists are also known as a **data collection**: data types that can store multiple data items.

```
nums = [0, 4, 6, 8, 11]     # declaring a list of numbers
```

List methods

Method	What it does	Example
append(x)	Adds an item to the end of an existing list	L = [4,12,3]; L.append(5) → [4,12,3,5]
insert(i, x)	Inserts an item at a given position (position first, then item)	L = [4,12,3]; L.insert(0,15) → [15,4,12,3]
remove(x)	Removes the first item whose value is x	L = [4,12,3,12]; L.remove(12) → [4,3,12]
pop(i)	Removes the item at position i and returns it; with no index, removes and returns the last item	L = [4,12,3,34,56]; L.pop(3) returns 34, L → [4,12,3,56]
clear()	Removes all items, leaving an empty list	L.clear() → []
len(L)	Returns the length (number of items) of the list	len([4,12,3,34,56]) → 5
index(x, start, end)	Returns the zero-based index of the first item whose value is x; raises ValueError if there is no such item. start/end limit the search like slice notation, but the returned index is relative to the start of the full list	
count(x)	Returns the number of times x appears in the list	
sort(key=None, reverse=False)	Sorts the items of the list in place	
reverse()	Reverses the items of the list in place	
copy()	Returns a shallow copy of the list	

Slicing

Get a sub-list with `list_variable[start : stop : step]`. Defaults: start = 0, stop = end of the list, step = 1.

- **start**: starting index position
- **stop**: slicing stops before this index
- **step**: take every n-th element between start and stop

```
Mylist = [17, 23, 12, 56, 45, 78, 55, 90, 11]
```

```
#  index   0   1   2   3   4   5   6   7   8

Mylist[0:]    -> [17, 23, 12, 56, 45, 78, 55, 90, 11]
Mylist[2:5]   -> [12, 56, 45]
Mylist[:4]    -> [17, 23, 12, 56]
Mylist[1:6]   -> [23, 12, 56, 45, 78]
Mylist[1:6:3] -> [23, 45]
Mylist[1:6:2] -> [23, 56, 78]
```